Supplementary material

FRESH Solutions: an adapter and exact-replay harness for metaheuristic benchmarking

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This supplement reports the per-instance results summarised in the main text. Gap values are percentages to the known optimum, rounded to three decimal places. Counts, seeds, and budgets are exact. The evidence files are `paper2/evidence/qap-expanded-100k-10seed.json`, `qap-bls-10seed.json`, `tsp-expanded-10k.json`, and `framework-use.json`. Every incumbent listed here replayed exactly through the corresponding problem adapter.

1 Framework-use panel

Four QAPLIB instances, ten seeds, a 100,000-iteration patience rule, and a 1,000,000-iteration cap. The CPU ratio is the framework-managed path divided by the private-array delta path.

Table 1: Framework-use panel. All 40 runs plateau-stopped, matched the standalone and production objectives, and replayed exactly.

Instance	Mean gap	Median gap	Worst gap	Mean iterations	CPU ratio	Patience
`tai12a`	0.000	0.000	0.000	100,186	1.33	10/10
`tai20a`	0.030	0.000	0.304	147,748	1.17	10/10
`tai25a`	0.077	0.000	0.402	175,611	1.18	10/10
`tai30a`	0.245	0.391	0.491	161,046	1.28	10/10

2 QAP panel: tabu reference and Breakout Local Search

Twenty QAPLIB instances, ten seeds per arm, a 100,000-iteration or 100,000-move patience rule, and a 1,000,000-unit cap. All 400 runs plateau-stopped and replayed exactly. BLS is a reimplementation of the published design, not the original authors' code.

Table 2: Per-instance QAP results. "Optima" counts runs at the known optimum.

		Tabu reference			BLS		
Instance	n	Mean	Worst	Optima	Mean	Worst	Optima
tai12a	12	0.000	0.000	10/10	0.000	0.000	10/10
tai20a	20	0.030	0.304	9/10	0.030	0.304	9/10
tai25a	25	0.077	0.402	8/10	0.000	0.000	10/10
tai30a	30	0.245	0.491	4/10	0.078	0.391	8/10
tai35a	35	0.599	1.051	1/10	0.252	0.649	3/10
tai40a	40	0.827	1.106	0/10	0.439	0.810	0/10
tai50a	50	1.170	1.583	0/10	0.739	1.013	0/10
bur26a	26	0.123	0.147	0/10	0.000	0.000	10/10
bur26b	26	0.565	0.565	0/10	0.000	0.000	10/10
bur26c	26	0.004	0.004	0/10	0.000	0.000	10/10
bur26d	26	0.041	0.043	0/10	0.000	0.000	10/10
bur26e	26	0.009	0.009	0/10	0.000	0.000	10/10
bur26f	26	0.383	0.542	0/10	0.000	0.000	10/10
bur26g	26	0.376	0.376	0/10	0.000	0.000	10/10
bur26h	26	0.368	0.368	0/10	0.000	0.000	10/10
chr12a	12	0.570	5.695	9/10	0.000	0.000	10/10
chr15a	15	0.000	0.000	10/10	0.000	0.000	10/10
chr18a	18	0.000	0.000	10/10	0.000	0.000	10/10
chr20a	20	0.630	1.460	2/10	0.228	1.369	4/10
chr22a	22	0.000	0.000	10/10	0.156	0.617	7/10

3 Tabu-reference diversity on the bur26 family

The eight asymmetric bur26 instances are not independent replicates. The tabu reference reaches no optimum in 80 runs; four instances produce a single gap value across all ten seeds, and the others produce two or three values. This is a property of the unvalidated reference, not evidence about the tabu reference in general.

Table 3: Distinct tabu-reference gap values across ten seeds per bur26 instance.

Instance	Distinct gaps	Best gap	Worst gap
bur26a	2	0.088	0.147
bur26b	2	0.565	0.565
bur26c	1	0.004	0.004
bur26d	3	0.035	0.043
bur26e	1	0.009	0.009
bur26f	2	0.014	0.542
bur26g	1	0.376	0.376
bur26h	1	0.368	0.368

4 TSP breadth panel

Twenty TSPLIB instances, five seeds, a 10,000-kick cap, a 1,000-kick patience rule, and a 3,600-second safety cap. The panel is frozen with the per-instance maximum exchange depth shown. It is not one adaptive-depth experiment. Five brazil58 runs reached the known optimum but hit the wall-clock cap and are reported as cap hits, not patience stops.

Table 4: Per-instance TSP breadth results. Mean wall time is seconds per run on the campaign host.

Instance	n	Depth	Mean gap	Worst gap	Patience	Optima	Mean wall (s)
att48	48	4	0.000	0.000	5/5	5/5	134
berlin52	52	4	0.000	0.000	5/5	5/5	637
bier127	127	4	0.000	0.000	5/5	5/5	2,180
brazil58	58	4	0.000	0.000	0/5	5/5	3,600
ch130	130	4	0.000	0.000	5/5	5/5	902
ch150	150	4	0.000	0.000	5/5	5/5	526
d198	198	2	0.005	0.006	5/5	1/5	895
eil51	51	4	0.000	0.000	5/5	5/5	38
eil76	76	4	0.000	0.000	5/5	5/5	49
eil101	101	4	0.000	0.000	5/5	5/5	149
fri26	26	4	0.000	0.000	5/5	5/5	67
gr120	120	4	0.000	0.000	5/5	5/5	1,700
pr152	152	2	0.074	0.185	5/5	3/5	269
kroA100	100	4	0.000	0.000	5/5	5/5	545
kroB100	100	4	0.000	0.000	5/5	5/5	559
kroC100	100	4	0.000	0.000	5/5	5/5	339
kroD100	100	4	0.000	0.000	5/5	5/5	395
kroE100	100	4	0.000	0.000	5/5	5/5	514
lin105	105	3	0.000	0.000	5/5	5/5	222
pr76	76	4	0.000	0.000	5/5	5/5	335